Ecuación Principal

$$F_a(t) = F_z(t) + F_L(t) = F_c(t) + F_R(t)$$

$$F_z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L} \int \left[P_a(t) - P_p(t) \right] dt$$

$$F_c(t) = C \frac{d P_p(t)}{dt} \qquad F_R(t) = \frac{P_p(t)}{R}$$

Procedimiento algebraico

$$\frac{P_a(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L} \int \left[P_a(t) P_p(t) \right] dt = \frac{C d P_p(t)}{dt} + \frac{P_p(t)}{R_1}$$

$$\frac{Pa(s)}{Z} - \frac{Pp(s)}{Z} + \frac{Pa(s) - Pp(s)}{LS} = CSP_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{LS}\right) Pa(s) = \left(CS + \frac{1}{R} + \frac{1}{Z} + \frac{1}{LS}\right) P_p(s)$$

$$\frac{LS+Z}{LZS} = \frac{CRLZS^2 + LZS + RLS + RZ}{RLZS} \, Pp(s)$$

$$\frac{Pp(s)}{Pa(s)} = \frac{\dfrac{LS+Z}{LZS}}{\dfrac{CLS^2 + LZS + RLS + RZ}{RLSZ}}$$

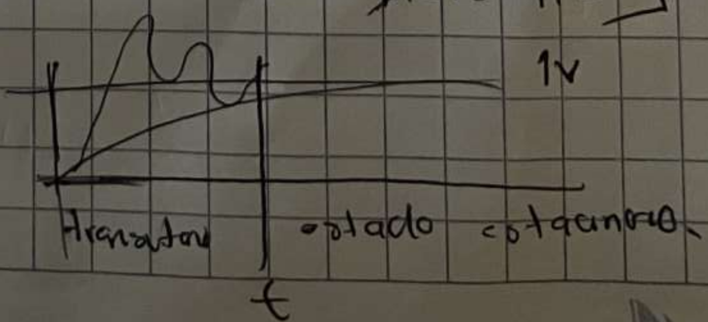
$$\frac{RLS + RZ}{CLRZS^2 + (LZ + RL)S + RL} \qquad (LS+Z)(RLSZ)$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s \, Pa(s) \left[1 - \frac{Pp(s)}{Pa(s)}\right]$$

$$= \lim_{s \to 0} \cancel{s} \cdot \frac{1}{\cancel{s}} \left[1 - \frac{RL\cancel{s} + RZ}{CL\cancel{RZ}\,s^2 + (LZ + \cancel{RL})s + RZ}\right]$$

$$= 1 - \frac{RZ}{RZ} = 0V$$



Estabilidad en Lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$a = CLRZ$

$b = LZ + RL$

$c = RZ$

$$\lambda_{1,2} = \frac{-(LZ+RL) \pm \sqrt{(LZ+RL)^2 - 4CLR^2Z^2}}{2CLRZ} = \frac{(-)(\,)}{+}$$

El sistema tiene una respuesta estable

Porque $Re\,\lambda_{1,2} < 0$

Modelo de ecuaciones integro-diferenciales

$$P_p(t)\Big(\overbrace{\frac{1}{R} + \frac{1}{Z}}^{\frac{Z+R}{ZR}}\Big) = \frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - \frac{C\,dP_p(t)}{dt}$$

$$P_p(t) = \left(\frac{P_a(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt - \frac{C\,dP_p(t)}{dt}\right)\frac{ZR}{Z+R}$$